Supplementary Material
SoK: How Hard Is the Attack? A Complexity-Theoretic Systematization of Machine Learning Security

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1. Classifying the corpus

This section reports the corpus in two layers. Tables 2 to 4 cross-tabulate features against one another, with works in the cells by citation number. Tables 5 and 6 give one row per work, so that every cell above is traceable to the paper behind it. Every value comes from a fixed vocabulary settled before the pass began.

The cross-tabulations are the layer that carries an argument, because in them an empty cell is a combination nobody has published. The value `none` is an answer and not a missing entry, recording that a work constrains no party at all. Of the 200 works, 37 state no resource bound on any party, and 0 fall outside the frame in the sense that deleting every resource restriction leaves the main claim unchanged.

TABLE 1: The corpus by pipeline phase (rows) against access level (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

phase	**hard label**	**scores**	**black box**	**white box**	**weights**	**gradients**	**channel**	**none**	***unassigned***	Σ
train time			[106], [147]	[69], [72], [138], [141], [167], [186], [196], [198]		[28]	[193]	[12], [22], [71], [73], [81], [161], [169], [191]		20
test time	[93]		[32], [58], [97], [98], [102], [159], [168], [175]	[6], [13], [49], [50], [55], [66], [67], [99], [144], [145], [174], [177], [178], [182], [185], [188]			[115]	[10], [40], [43], [51], [79], [86], [94], [109], [137], [143], [151], [152], [157], [171], [200]		41
post hoc		[16], [53], [54], [62], [87], [118], [123], [160], [173], [181], [197]	[52], [75], [110], [125], [153], [172], [189]	[25], [59], [60], [96], [101], [127], [131], [135], [146], [158], [164], [183], [187], [194]	[122]	[128]		[64], [91], [150], [154], [180]	[165]	40
n/a	[63]	[80]	[78]					[1], [2], [3], [4], [5], [7], [8], [9], [11], [14], [15], [17], [18], [19], [20], [21], [23], [24], [26], [27], [29], [30], [31], [33], [34], [35], [36], [37], [38], [39], [41], [42], [44], [45], [46], [47], [48], [56], [57], [61], [65], [68], [70], [74], [76], [77], [82], [83], [84], [85], [88], [89], [90], [92], [95], [100], [103], [104], [105], [107], [108], [111], [112], [113], [114], [116], [117], [119], [120], [121], [124], [126], [129], [130], [132], [133], [134], [136], [139], [140], [142], [148], [149], [155], [156], [162], [163], [166], [170], [176], [179], [184], [190], [192], [195], [199]		99
Σ	2	12	18	38	1	2	2	124	1	200

TABLE 2: The corpus by constrained party (rows) against resource unit (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies. A dagger marks works where the resource appears only as a side condition rather than as a parameter the work assesses on purpose.

party	time	circuit size	queries	samples	memory	algo class	parameters	perturbation	corruption	rounds	privacy budget	circuit depth	success p
attacker	[11], [26], [33], [45], [57], [63], [64], [72], [75], [80], [93], [101], [107], [118], [119], [127], [139], [148], [149], [181], [186], [193], [28], [54], [96], [175], [196], [198]†	[1], [23], [38], [48], [100], [102]	[5], [42], [52], [58], [123], [146], [53], [62], [97], [153], [173]†	[122], [165]†	[24], [84], [85], [111]	[61], [90], [125], [190]		[6], [13], [50], [66], [67], [79], [94], [115], [137], [152], [157], [171], [174], [177], [178], [182], [185], [188], [49], [55]†	[138], [141], [147], [161], [169]			[108]	
learner	[15], [19], [20], [31], [34], [36], [37], [40], [44], [47], [70], [73], [77], [81], [120], [124], [136], [142], [166], [179], [199]	[105]	[22]†	[7], [10], [29], [51], [71], [74], [76], [83], [89], [112], [126], [129], [140], [144], [145], [184], [191]	[163]	[2], [3], [8], [17], [21], [35], [43], [56], [68], [92], [95], [104], [117], [132], [176], [192]	[46], [103], [116]						
auditor	[59], [69], [106], [110], [131], [32], [197]†			[154], [200]		[78]							
mechanism none	[27], [162]				[88], [134]	[195]†					[4], [164]†		
Σ	58	7	12	21	7	22	3	20	5	0	2	1	0

TABLE 3: The corpus by resource unit (rows) against tightness of the bound (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

unit	unstated	poly	quasi-poly	class	budget	conditional	exponent	relative	measured	Σ
time	[197]	[27], [37], [40], [57], [59], [63], [64], [69], [73], [80], [81], [93], [101], [106], [107], [110], [118], [119], [120], [124], [131], [142], [148], [149], [179], [181], [186], [193], [199]	[34]		[162], [198]	[11], [15], [20], [36], [44], [70]	[47], [77], [136]	[26], [32], [33], [45], [127], [139], [196]	[19], [28], [31], [54], [72], [75], [96], [166], [175]	58
circuit size		[102]	[1], [48], [100]		[38]	[105]		[23]		7
queries					[22], [52], [97], [146], [153]			[5], [42], [123]	[53], [58], [62], [173]	12
samples				[71], [191]	[10], [51], [144], [145], [200]		[7], [29], [74], [76], [83], [89], [112], [126], [129], [140], [184]	[122], [154]	[165]	21
memory						[88]	[134], [163]	[24], [84], [85], [111]		7
algo class				[2], [3], [8], [17], [21], [35], [43], [56], [61], [68], [78], [90], [92], [95], [104], [117], [125], [132], [176], [190], [192], [195]					22
parameters					[46], [103], [116]					3
perturbation					[6], [13], [49], [50], [55], [66], [67], [79], [94], [115], [137], [152], [157], [171], [174], [177], [178], [182], [185], [188]					20
corruption					[138], [141], [147], [169]				[161]	5
rounds										0
privacy budget					[4], [164]					2
circuit depth								[108]		1
success prob										0
flops							[18], [114], [133], [156]		[113]	5
none	[9], [12], [14], [16], [25], [30], [39], [41], [60], [65], [82], [86], [87], [91], [98], [99], [109], [121], [128], [130], [135], [143], [150], [151], [155], [158], [159], [160], [167], [168], [170], [172], [180], [183], [187], [189], [194]									37
Σ	38	30	4	24	42	8	20	18	16	200

TABLE 4: The corpus by attack family (rows) against constrained party (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

family	**attacker**	**learner**	**auditor**	**mechanism**	**none**	Σ
membership	[54], [101], [122], [173], [186]		[197]		[16], [87], [91], [150], [160], [194]	12
attribute						0
reconstruct	[28], [64], [80], [118], [123], [146]				[25], [60], [135], [189]	10
distribution	[165], [181], [193]					3
extract	[52], [53], [62], [75], [125]				[172]	6
invert						0
forge	[72], [96], [127], [198]				[128], [167], [183]	7
evade	[6], [13], [49], [50], [55], [66], [67], [79], [93], [94], [97], [102], [115], [137], [152], [157], [171], [174], [175], [177], [178], [182], [185], [188]	[10], [40], [43], [51], [81], [144], [145]	[32]		[98], [99], [109], [143], [151]	37
evade audit	[58], [196]		[154]		[159], [180]	5
inject			[200]		[86]	2
poison	[138], [141], [147], [169]	[22], [71], [73], [191]				8
backdoor	[153], [161]		[59], [69], [106], [110], [131]		[12], [158], [187]	10
multiple				[164]	[130], [155]	3
none	[1], [5], [11], [23], [24], [26], [33], [38], [42], [45], [48], [57], [61], [63], [84], [85], [90], [100], [107], [108], [111], [119], [139], [148], [149], [190]	[2], [3], [7], [8], [15], [17], [18], [19], [20], [21], [29], [31], [34], [35], [36], [37], [44], [46], [47], [56], [68], [70], [74], [76], [77], [83], [89], [92], [95], [103], [104], [105], [112], [113], [114], [116], [117], [120], [124], [126], [129], [132], [133], [136], [140], [142], [156], [163], [166], [176], [179], [184], [192], [199]	[78]	[4], [27], [88], [134], [162], [195]	[9], [14], [30], [39], [41], [65], [82], [121], [168], [170]	97
Σ	81	65	10	7	37	200

TABLE 5: Per-work reference layer: attack scenario and threat model.

#	Target	Family	Phase	Access	Write	Frame
[91]	train data	membership	post hoc	none	read only	none
[160]	train data	membership	post hoc	scores	read only	none
[87]	train data	membership	post hoc	scores	read only	none
[186]	train data	membership	train time	white box	corpus	compute
[122]	train data	membership	post hoc	weights	read only	statistical
[173]	train data	membership	post hoc	scores	read only	compute
[197]	train data	membership	post hoc	scores	read only	compute
[54]	train data	membership	post hoc	scores	read only	compute
[16]	train data	membership	post hoc	scores	read only	none
[194]	train data	membership	post hoc	white box	read only	none
[150]	train data	membership	post hoc	none	none	none
[101]	train data	membership	post hoc	white box	read only	compute
[60]	train data	reconstruct	post hoc	white box	read only	none
[64]	train data	reconstruct	post hoc	none	read only	compute
[80]	train data	reconstruct	n/a	scores	read only	compute
[135]	train data	reconstruct	post hoc	white box	read only	none
[28]	train data	reconstruct	train time	gradients	read only	compute
[123]	train data	reconstruct	post hoc	scores	read only	compute
[146]	train data	reconstruct	post hoc	white box	read only	compute
[25]	train data	reconstruct	post hoc	white box	read only	none
[189]	train data	reconstruct	post hoc	black box	read only	none
[118]	train data	reconstruct	post hoc	scores	read only	compute
[165]	train data	distribution	post hoc	·	read only	statistical
[181]	train data	distribution	post hoc	scores	read only	compute
[193]	train data	distribution	train time	channel	corpus	compute
[52]	parameters	extract	post hoc	black box	read only	compute
[172]	parameters	extract	post hoc	black box	read only	none
[75]	parameters	extract	post hoc	black box	read only	compute
[62]	parameters	extract	post hoc	scores	read only	compute
[125]	parameters	extract	post hoc	black box	read only	compute
[53]	parameters	extract	post hoc	scores	read only	compute
[183]	training run	forge	post hoc	white box	corpus	none
[128]	train data	forge	post hoc	gradients	corpus	none
[198]	training run	forge	train time	white box	model	compute
[96]	training run	forge	post hoc	white box	corpus	compute
[72]	training run	forge	train time	white box	model	compute
[127]	training run	forge	post hoc	white box	model	compute
[167]	training run	forge	train time	white box	corpus	none
[182]	outputs	evade	test time	white box	read only	capability
[55]	outputs	evade	test time	white box	read only	capability
[171]	outputs	evade	test time	none	read only	capability
[43]	outputs	evade	test time	none	read only	compute
[40]	outputs	evade	test time	none	none	compute
[109]	outputs	evade	test time	none	read only	none
[94]	outputs	evade	test time	none	read only	capability
[10]	outputs	evade	test time	none	read only	statistical
[137]	outputs	evade	test time	none	read only	capability
[66]	outputs	evade	test time	white box	read only	capability
[152]	outputs	evade	test time	none	read only	capability
[93]	outputs	evade	test time	hard label	read only	compute
[102]	outputs	evade	test time	black box	read only	compute
[143]	outputs	evade	test time	none	read only	none
[157]	outputs	evade	test time	none	read only	capability
[151]	outputs	evade	test time	none	read only	none
[32]	outputs	evade	test time	black box	read only	compute
[99]	outputs	evade	test time	white box	read only	none
[98]	outputs	evade	test time	black box	read only	none
[79]	outputs	evade	test time	none	read only	capability

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[67]	outputs	evade	test time	white box	read only	capability
[6]	outputs	evade	test time	white box	read only	capability
[49]	outputs	evade	test time	white box	read only	capability
[51]	outputs	evade	test time	none	read only	statistical
[188]	outputs	evade	test time	white box	read only	capability
[178]	outputs	evade	test time	white box	read only	capability
[174]	outputs	evade	test time	white box	read only	capability
[145]	outputs	evade	test time	white box	read only	statistical
[50]	outputs	evade	test time	white box	read only	capability
[185]	outputs	evade	test time	white box	read only	capability
[115]	outputs	evade	test time	channel	channel	capability
[177]	outputs	evade	test time	white box	read only	capability
[97]	outputs	evade	test time	black box	read only	compute
[175]	outputs	evade	test time	black box	read only	compute
[13]	outputs	evade	test time	white box	read only	capability
[144]	outputs	evade	test time	white box	none	statistical
[81]	none	evade	train time	none	none	compute
[180]	training run	evade audit	post hoc	none	none	none
[154]	training run	evade audit	post hoc	none	responses	statistical
[159]	outputs	evade audit	test time	black box	read only	none
[58]	outputs	evade audit	test time	black box	responses	compute
[196]	training run	evade audit	train time	white box	model	compute
[86]	outputs	inject	test time	none	responses	none
[200]	outputs	inject	test time	none	channel	statistical
[169]	train data	poison	train time	none	corpus	capability
[141]	train data	poison	train time	white box	corpus	capability
[138]	train data	poison	train time	white box	corpus	capability
[147]	train data	poison	train time	black box	corpus	capability
[73]	train data	poison	train time	none	corpus	compute
[71]	train data	poison	train time	none	corpus	compute
[191]	train data	poison	train time	none	corpus	compute
[22]	train data	poison	train time	none	corpus	compute
[59]	parameters	backdoor	post hoc	white box	read only	compute
[158]	parameters	backdoor	post hoc	white box	read only	none
[153]	outputs	backdoor	post hoc	black box	read only	compute
[187]	parameters	backdoor	post hoc	white box	read only	none
[131]	parameters	backdoor	post hoc	white box	model	compute
[110]	parameters	backdoor	post hoc	black box	read only	compute
[106]	parameters	backdoor	train time	black box	model	compute
[12]	train data	backdoor	train time	none	corpus	none
[69]	parameters	backdoor	train time	white box	model	compute
[161]	parameters	backdoor	train time	none	corpus	capability
[164]	train data	multiple	post hoc	white box	read only	capability
[130]	none	multiple	n/a	none	none	none
[155]	none	multiple	n/a	none	none	none
[134]	none	none	n/a	none	none	compute
[113]	none	none	n/a	none	none	compute
[9]	none	none	n/a	none	none	none
[84]	none	none	n/a	none	none	compute
[45]	none	none	n/a	none	none	compute
[77]	none	none	n/a	none	none	compute
[17]	none	none	n/a	none	none	compute
[35]	none	none	n/a	none	none	compute
[38]	none	none	n/a	none	none	compute
[111]	none	none	n/a	none	none	compute
[21]	none	none	n/a	none	none	compute
[44]	none	none	n/a	none	none	compute
[24]	none	none	n/a	none	none	compute
[61]	train data	none	n/a	none	none	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[119]	none	none	n/a	none	none	compute
[149]	none	none	n/a	none	none	compute
[114]	none	none	n/a	none	none	compute
[30]	none	none	n/a	none	none	none
[148]	train data	none	n/a	none	none	compute
[156]	none	none	n/a	none	none	compute
[63]	training run	none	n/a	hard label	read only	compute
[1]	none	none	n/a	none	none	compute
[166]	none	none	n/a	none	none	compute
[4]	train data	none	n/a	none	none	capability
[37]	none	none	n/a	none	none	compute
[70]	none	none	n/a	none	none	compute
[47]	none	none	n/a	none	none	compute
[14]	none	none	n/a	none	none	none
[88]	none	none	n/a	none	none	compute
[162]	train data	none	n/a	none	none	compute
[18]	none	none	n/a	none	none	compute
[29]	none	none	n/a	none	none	compute
[19]	none	none	n/a	none	none	compute
[95]	none	none	n/a	none	none	compute
[90]	none	none	n/a	none	none	compute
[36]	none	none	n/a	none	none	compute
[15]	none	none	n/a	none	none	compute
[5]	none	none	n/a	none	none	compute
[26]	none	none	n/a	none	none	compute
[23]	none	none	n/a	none	none	compute
[11]	none	none	n/a	none	none	compute
[168]	outputs	none	test time	black box	corpus	none
[129]	none	none	n/a	none	none	compute
[105]	none	none	n/a	none	none	compute
[104]	none	none	n/a	none	none	compute
[83]	none	none	n/a	none	none	compute
[76]	none	none	n/a	none	none	compute
[142]	none	none	n/a	none	none	compute
[179]	none	none	n/a	none	none	compute
[31]	none	none	n/a	none	none	compute
[121]	none	none	n/a	none	none	none
[124]	none	none	n/a	none	none	compute
[195]	none	none	n/a	none	none	compute
[8]	none	none	n/a	none	none	compute
[78]	outputs	none	n/a	black box	read only	compute
[39]	none	none	n/a	none	none	none
[92]	none	none	n/a	none	none	compute
[170]	none	none	n/a	none	none	none
[3]	none	none	n/a	none	none	compute
[68]	none	none	n/a	none	none	compute
[120]	none	none	n/a	none	none	compute
[184]	none	none	n/a	none	none	compute
[107]	none	none	n/a	none	none	compute
[56]	none	none	n/a	none	none	compute
[176]	none	none	n/a	none	none	compute
[192]	none	none	n/a	none	none	compute
[65]	none	none	n/a	none	none	none
[2]	none	none	n/a	none	none	compute
[42]	none	none	n/a	none	none	compute
[190]	none	none	n/a	none	none	compute
[74]	none	none	n/a	none	none	compute
[103]	none	none	n/a	none	none	compute
[33]	none	none	n/a	none	none	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[139]	none	none	n/a	none	none	compute
[48]	none	none	n/a	none	none	compute
[100]	none	none	n/a	none	none	compute
[136]	none	none	n/a	none	none	compute
[34]	none	none	n/a	none	none	compute
[57]	none	none	n/a	none	none	compute
[163]	none	none	n/a	none	none	compute
[20]	none	none	n/a	none	none	compute
[7]	none	none	n/a	none	none	compute
[89]	none	none	n/a	none	none	compute
[117]	none	none	n/a	none	none	compute
[133]	none	none	n/a	none	none	compute
[27]	none	none	n/a	none	none	compute
[108]	none	none	n/a	none	none	compute
[140]	none	none	n/a	none	none	compute
[112]	none	none	n/a	none	none	compute
[126]	none	none	n/a	none	none	compute
[82]	none	none	n/a	none	none	none
[85]	none	none	n/a	none	none	compute
[116]	none	none	n/a	none	none	compute
[46]	none	none	n/a	none	none	compute
[132]	train data	none	n/a	none	none	compute
[199]	none	none	n/a	none	none	compute
[41]	none	none	n/a	none	none	none

TABLE 6: Per-work reference layer: adversary assumptions and claim shape. Same rows and same order as Table 5.

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[91]	none	none	unstated	stated	none	none	·	empirical	stated
[160]	none	none	unstated	stated	none	none	·	empirical	stated
[87]	none	none	unstated	stated	none	none	·	empirical	stated
[186]	attacker	time	poly	stated	none	none	explicit	conditional	stated
[122]	attacker	samples	relative	absent	samples	·	explicit	universal	stated
[173]	attacker	queries	measured	stated	none	none	implicit	empirical	stated
[197]	auditor	time	unstated	absent	none	none	implicit	conditional	stated
[54]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[16]	none	none	unstated	absent	none	none	·	empirical	stated
[194]	none	none	unstated	stated	none	none	·	conditional	stated
[150]	none	none	unstated	stated	none	none	·	empirical	stated
[101]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[60]	none	none	unstated	stated	none	·	·	universal	stated
[64]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[80]	attacker	time	poly	absent	perturbation	queries	explicit	universal	stated
[135]	none	none	unstated	stated	none	·	·	existential	stated
[28]	attacker	time	measured	absent	none	none	implicit	empirical	stated
[123]	attacker	queries	relative	absent	samples	none	explicit	conditional	stated
[146]	attacker	queries	budget	absent	privacy budget	none	explicit	universal	stated
[25]	none	none	unstated	stated	none	none	·	conditional	stated
[189]	none	none	unstated	n/a	none	none	·	empirical	stated
[118]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[165]	attacker	samples	measured	stated	none	none	implicit	empirical	stated
[181]	attacker	time	poly	stated	·	none	explicit	conditional	stated
[193]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[52]	attacker	queries	budget	absent	none	none	explicit	conditional	stated
[172]	none	none	unstated	absent	none	none	·	empirical	stated
[75]	attacker	time	measured	absent	none	none	explicit	empirical	stated
[62]	attacker	queries	measured	absent	none	none	implicit	empirical	stated
[125]	attacker	algo class	class	stated	none	none	explicit	existential	stated
[53]	attacker	queries	measured	absent	none	time	implicit	empirical	stated
[183]	none	none	unstated	n/a	none	none	·	existential	stated
[128]	none	none	unstated	absent	none	none	·	empirical	stated
[198]	attacker	time	budget	n/a	time	none	implicit	conditional	stated
[96]	attacker	time	measured	absent	none	none	implicit	empirical	stated
[72]	attacker	time	measured	absent	none	none	explicit	empirical	stated
[127]	attacker	time	relative	absent	time	none	explicit	conditional	stated
[167]	none	none	unstated	n/a	none	none	·	empirical	stated
[182]	attacker	perturbation	budget	absent	none	none	explicit	existential	none
[55]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[171]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[43]	learner	algo class	class	n/a	perturbation	·	explicit	existential	stated
[40]	learner	time	poly	stated	perturbation	none	explicit	conditional	stated
[109]	none	none	unstated	n/a	none	none	·	existential	stated
[94]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[10]	learner	samples	budget	n/a	none	·	explicit	universal	stated
[137]	attacker	perturbation	budget	n/a	none	·	explicit	universal	stated
[66]	attacker	perturbation	budget	n/a	none	none	explicit	universal	stated
[152]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[93]	attacker	time	poly	n/a	none	none	explicit	universal	stated
[102]	attacker	circuit size	poly	n/a	none	none	explicit	existential	stated
[143]	none	none	unstated	n/a	none	none	·	conditional	stated
[157]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[151]	none	none	unstated	n/a	none	none	·	empirical	stated
[32]	auditor	time	relative	stated	time	·	implicit	existential	stated
[99]	none	none	unstated	n/a	none	·	·	conditional	stated
[98]	none	none	unstated	n/a	none	none	·	empirical	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[79]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[67]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[6]	attacker	perturbation	budget	absent	none	none	explicit	existential	none
[49]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[51]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[188]	attacker	perturbation	budget	n/a	none	none	explicit	universal	stated
[178]	attacker	perturbation	budget	n/a	none	none	explicit	empirical	stated
[174]	attacker	perturbation	budget	n/a	samples	none	explicit	existential	stated
[145]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[50]	attacker	perturbation	budget	stated	none	time	explicit	empirical	stated
[185]	attacker	perturbation	budget	stated	none	none	explicit	existential	stated
[115]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[177]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[97]	attacker	queries	budget	stated	none	algo class	implicit	existential	none
[175]	attacker	time	measured	n/a	none	none	implicit	empirical	none
[13]	attacker	perturbation	budget	n/a	samples	none	explicit	universal	stated
[144]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[81]	learner	time	poly	n/a	perturbation	none	explicit	conditional	stated
[180]	none	none	unstated	n/a	none	·	·	conditional	stated
[154]	auditor	samples	relative	n/a	samples	none	explicit	conditional	stated
[159]	none	none	unstated	n/a	none	none	·	existential	stated
[58]	attacker	queries	measured	stated	none	time	explicit	existential	stated
[196]	attacker	time	relative	n/a	time	·	implicit	existential	stated
[86]	none	none	unstated	n/a	none	none	·	empirical	stated
[200]	auditor	samples	budget	n/a	none	none	explicit	conditional	stated
[169]	attacker	corruption	budget	absent	algo class	none	explicit	conditional	stated
[141]	attacker	corruption	budget	n/a	none	none	explicit	conditional	stated
[138]	attacker	corruption	budget	n/a	none	·	explicit	conditional	stated
[147]	attacker	corruption	budget	stated	none	perturbation	explicit	universal	stated
[73]	learner	time	poly	n/a	corruption	·	explicit	existential	stated
[71]	learner	samples	class	n/a	none	algo class	explicit	conditional	stated
[191]	learner	samples	class	n/a	none	algo class	explicit	universal	stated
[22]	learner	queries	budget	n/a	corruption	time	implicit	universal	stated
[59]	auditor	time	poly	stated	corruption	none	explicit	conditional	stated
[158]	none	none	unstated	n/a	none	·	·	existential	none
[153]	attacker	queries	budget	n/a	none	time	implicit	empirical	stated
[187]	none	none	unstated	n/a	none	none	·	existential	none
[131]	auditor	time	poly	stated	perturbation	none	explicit	existential	stated
[110]	auditor	time	poly	absent	corruption	samples	explicit	conditional	stated
[106]	auditor	time	poly	absent	none	none	explicit	existential	stated
[12]	none	none	unstated	n/a	none	none	·	empirical	none
[69]	auditor	time	poly	n/a	none	none	explicit	empirical	stated
[161]	attacker	corruption	measured	stated	none	·	explicit	empirical	stated
[164]	mechanism	privacy budget	budget	stated	none	none	implicit	universal	stated
[130]	none	none	unstated	n/a	none	none	·	empirical	stated
[155]	none	none	unstated	n/a	none	none	·	empirical	stated
[134]	mechanism	memory	exponent	n/a	none	·	explicit	existential	stated
[113]	learner	flops	measured	n/a	none	none	explicit	empirical	stated
[9]	none	none	unstated	n/a	none	none	·	conditional	stated
[84]	attacker	memory	relative	absent	memory	none	explicit	universal	stated
[45]	attacker	time	relative	n/a	time	none	explicit	conditional	stated
[77]	learner	time	exponent	n/a	none	none	explicit	conditional	stated
[17]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[35]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[38]	attacker	circuit size	budget	n/a	circuit size	algo class	explicit	existential	stated
[111]	attacker	memory	relative	n/a	memory	queries	explicit	universal	stated
[21]	learner	algo class	class	n/a	none	algo class	explicit	conditional	stated
[44]	learner	time	conditional	n/a	none	algo class	explicit	conditional	stated
[24]	attacker	memory	relative	n/a	memory	time	explicit	conditional	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[61]	attacker	algo class	class	absent	none	·	explicit	existential	stated
[119]	attacker	time	poly	n/a	none	none	explicit	conditional	stated
[149]	attacker	time	poly	n/a	none	none	explicit	conditional	stated
[114]	learner	flops	exponent	n/a	none	parameters	explicit	empirical	stated
[30]	none	none	unstated	n/a	none	none	·	empirical	stated
[148]	attacker	time	poly	absent	none	none	explicit	existential	stated
[156]	learner	flops	exponent	n/a	none	parameters	explicit	conditional	stated
[63]	attacker	time	poly	stated	none	·	explicit	conditional	stated
[1]	attacker	circuit size	quasi-poly	stated	none	none	explicit	conditional	stated
[166]	learner	time	measured	n/a	none	parameters	explicit	conditional	stated
[4]	mechanism	privacy budget	budget	n/a	none	none	explicit	empirical	stated
[37]	learner	time	poly	n/a	none	·	explicit	conditional	stated
[70]	learner	time	conditional	stated	none	algo class	explicit	conditional	stated
[47]	learner	time	exponent	n/a	none	parameters	explicit	empirical	stated
[14]	none	none	unstated	n/a	none	none	·	empirical	none
[88]	mechanism	memory	conditional	n/a	time	none	explicit	conditional	stated
[162]	mechanism	time	budget	absent	none	·	explicit	universal	stated
[18]	learner	flops	exponent	n/a	none	parameters	explicit	conditional	stated
[29]	learner	samples	exponent	n/a	none	parameters	explicit	conditional	stated
[19]	learner	time	measured	n/a	none	none	explicit	conditional	stated
[95]	learner	algo class	class	n/a	none	queries	explicit	universal	stated
[90]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[36]	learner	time	conditional	n/a	none	·	explicit	conditional	stated
[15]	learner	time	conditional	n/a	none	none	explicit	conditional	stated
[5]	attacker	queries	relative	stated	queries	none	explicit	existential	stated
[26]	attacker	time	relative	stated	time	none	explicit	conditional	stated
[23]	attacker	circuit size	relative	n/a	circuit size	none	explicit	existential	stated
[11]	attacker	time	conditional	n/a	none	none	explicit	conditional	stated
[168]	none	none	unstated	n/a	none	none	·	empirical	stated
[129]	learner	samples	exponent	n/a	none	time	explicit	universal	stated
[105]	learner	circuit size	conditional	n/a	none	none	explicit	universal	stated
[104]	learner	algo class	class	n/a	none	·	explicit	universal	stated
[83]	learner	samples	exponent	n/a	none	algo class	explicit	conditional	stated
[76]	learner	samples	exponent	n/a	none	algo class	explicit	conditional	stated
[142]	learner	time	poly	stated	none	·	explicit	conditional	stated
[179]	learner	time	poly	stated	perturbation	·	explicit	conditional	stated
[31]	learner	time	measured	n/a	none	samples	explicit	empirical	stated
[121]	none	none	unstated	n/a	none	·	·	universal	stated
[124]	learner	time	poly	n/a	none	algo class	explicit	existential	stated
[195]	mechanism	algo class	class	n/a	none	time	implicit	conditional	stated
[8]	learner	algo class	class	n/a	none	samples	explicit	universal	stated
[78]	auditor	algo class	class	n/a	privacy budget	samples	explicit	existential	stated
[39]	none	none	unstated	n/a	none	none	·	conditional	stated
[92]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[170]	none	none	unstated	n/a	none	none	·	universal	stated
[3]	learner	algo class	class	n/a	none	time	explicit	conditional	stated
[68]	learner	algo class	class	n/a	none	queries	explicit	universal	stated
[120]	learner	time	poly	n/a	none	none	explicit	conditional	stated
[184]	learner	samples	exponent	n/a	none	none	explicit	conditional	stated
[107]	attacker	time	poly	n/a	privacy budget	none	explicit	conditional	stated
[56]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[176]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[192]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[65]	none	none	unstated	n/a	none	none	·	conditional	stated
[2]	learner	algo class	class	n/a	none	samples	explicit	existential	stated
[42]	attacker	queries	relative	n/a	queries	none	explicit	universal	stated
[190]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[74]	learner	samples	exponent	n/a	none	algo class	explicit	conditional	stated
[103]	learner	parameters	budget	n/a	none	time	explicit	existential	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[33]	attacker	time	relative	n/a	time	none	explicit	conditional	stated
[139]	attacker	time	relative	n/a	time	none	explicit	conditional	stated
[48]	attacker	circuit size	quasi-poly	stated	none	none	explicit	conditional	stated
[100]	attacker	circuit size	quasi-poly	stated	none	none	explicit	conditional	stated
[136]	learner	time	exponent	n/a	none	success prob	explicit	conditional	stated
[34]	learner	time	quasi-poly	n/a	none	algo class	explicit	existential	stated
[57]	attacker	time	poly	stated	none	none	explicit	existential	stated
[163]	learner	memory	exponent	n/a	none	none	explicit	universal	stated
[20]	learner	time	conditional	n/a	corruption	algo class	explicit	conditional	stated
[7]	learner	samples	exponent	n/a	none	algo class	explicit	universal	stated
[89]	learner	samples	exponent	n/a	none	time	explicit	universal	stated
[117]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[133]	learner	flops	exponent	n/a	none	parameters	explicit	empirical	stated
[27]	mechanism	time	poly	n/a	time	none	explicit	conditional	stated
[108]	attacker	circuit depth	relative	n/a	circuit depth	queries	explicit	existential	stated
[140]	learner	samples	exponent	n/a	none	algo class	explicit	universal	stated
[112]	learner	samples	exponent	n/a	none	algo class	explicit	universal	stated
[126]	learner	samples	exponent	n/a	none	time	explicit	conditional	stated
[82]	none	none	unstated	n/a	none	none	·	universal	stated
[85]	attacker	memory	relative	stated	memory	·	explicit	universal	stated
[116]	learner	parameters	budget	n/a	none	samples	explicit	universal	stated
[46]	learner	parameters	budget	n/a	none	none	explicit	conditional	stated
[132]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[199]	learner	time	poly	n/a	none	samples	explicit	existential	stated
[41]	none	none	unstated	n/a	none	none	·	empirical	stated

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